



Boron and phosphorous co-doped porous carbon as high-performance anode for sodium-ion battery

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ABSTRACT

Sodium-ion batteries (SIBs) have attracted extensive attention as the important replacement for lithium-ion batteries, due to the nature abundance of sodium sources. The key to high-performance SIBs lies in appropriate anode material with sufficient space and sites for the diffusion and adsorption of sodium ion (Na^+). Heteroatom doping in carbon has proven to be an effective strategy to improve the electrochemical performance of carbon-based anodes for SIB. The feasible preparation of doped carbon is essential for the development of SIBs. Here, boron (B) and phosphorous (P) co-doped honeycomb-like carbon (BPC) has been synthesized by one-step pyrolysis of onium salts containing B and P. Benefiting from dual doping of B and P in carbon, the increased layer space, defects and electrical conductivity of BPC enhance the adsorption capability of Na^+ , mass transport and charge diffusion. The formed porous structure in BPC can promote the electrolyte penetration as well as buffer the volume changes during cycling. When employed as the anode material for SIBs, high storage capacity and excellent cycle life have been enabled. This contribution paves a feasible and controlled approach to prepared heteroatom-doped carbons for Na^+ storage.

1. Introduction

- With the fast-growing demand of energy storage for renewable sources and power grids, the conventional lithium-ion batteries (LIBs) can hardly meet such desire, as the limited availability and consequently increasing cost of lithium source. Hence, sodium-ion batteries (SIBs) are broadly regarded as one of the most promising alternatives for lithium-ion batteries (LIBs), due to the natural abundance and environmental-friendliness of the sodium resource, [1–5]. The LIBs and SIBs share similar battery constitution and charge storage mechanism, especially regarding to the cathode. However, the transition from LIBs to SIBs have always been rough [6]. Sodium has greater atom mass (Na : 23 g mol^{-1} ; Li : 6.9 g mol^{-1}), and higher standard electrode potential (Na : -2.71 V vs. SHE; Li : -3.04 V vs. SHE), as compared to those of lithium. Also, the larger radius of Na^+ ($1.02 \text{ \AA}$), compared to Li^+ ($0.76 \text{ \AA}$), retards the ion diffusion rate and induces the sluggish kinetic process of SIBs during the charging/discharging process. These issues could further cause the low capacity and poor cycling performance of SIBs [7,8]. Hence, tremendous efforts have been devoted to exploring suitable

anode materials with high Na^+ storage capacity and rapid reaction dynamics to target high performance SIBs. The current pool of the anode materials for SIBs include carbonaceous material, metal oxide/sulfides and transition metal dichalcogenides [9,10]. Among them, carbon-based anode material, like graphene, carbon spheres, carbon skeletons, and hard carbon, have been firstly investigated as the anodes of SIBs, due to the good conductivity, low cost and resource wealth [6,7,11,12]. Unfortunately, most carbonaceous materials deliver deficient Na^+ storage capacity, particularly at the high current density. Accordingly, dopant and structure engineering are regarded as effective strategies to overcome those obstacles.

The dopant engineering, involved with the incorporation of single heteroatom like B, N, P, and S to carbon material, can tune the surface chemistry and tailor the structure of the carbon host. These dopants tend to improve electrical conductivity, layer distance, hence accelerating the charge transport as well as providing additional binding sites for Na^+ adsorption. Hence, heteroatom doping of carbons have been popularized to achieve SIBs with improved performance. Specifically, the N atom exhibits higher electronegativity than C atom,

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leading to the enhanced affinity for Na^+ and consequently excellent Na^+ absorption capability. Also, the larger ionic radius of sulfur and phosphorus (S: 102 pm, P: 110 pm, C: 77 pm), are beneficial for expanding the interlayer of carbon skeletons. The enlarged layer space can empower the accelerated Na^+ insertion/extraction and increase the storage capacity of SIBs [13–19]. Besides, boron is commonly used to replace carbon atoms, as the atomic radius of boron (B: 82 pm) is close to carbon. Hence, B can be easily incorporated into the carbon lattice, thereby increasing local defects in the carbon plane. Qiu et.al reported the introduction of boron at the trigonal spots in the carbon lattice, can function as the electron acceptor to increase the conductivity by shifting the Fermi level to conduction band [20]. Single heteroatom-doped carbon anodes have been proved highly effective in harvesting the high-performance of SIBs, like B-doped porous carbon (achieving 200 mAh g^{-1} after 100 cycles at 0.1 A g^{-1}) [8], N-doped porous carbon nanofibers (174 mAh g^{-1} after 300 cycles at 50 mA g^{-1}) [21] and S-doped graphene network (215 mAh g^{-1} after 100 cycles at 0.5C) [22]. In line with this, dual-dopant into porous carbons can further improve the behavior of the SIB compared to single heteroatom doping. For instance, the capacity of N,S co-doped porous carbon sheets achieved 223 mAh g^{-1} at 0.2 A g^{-1} after 500 cycles [23], the capacity of N/P co-doped porous carbon was 230 mAh g^{-1} at 0.2 A g^{-1} and 113 mAh g^{-1} at 10 A g^{-1} [18], B/N co-doped carbon materials offered the capacity as high as 329.71 mAh g^{-1} after 100 cycles at 0.1 A g^{-1} [24].

Besides, customized three-dimensional (3D) porous carbons with well-distributed pores, are particularly desired to obtain high-rate carbon-based anodes for SIBs. 3D porous structure can increase the contact area between the electrode and electrolyte to provide sufficient space for the redox reaction. Moreover, 3D structure can buffer volume change during insertion/extraction process, ensuring the structural integrity especially during cycling. Together with the shortened diffusion path of charges enabled by the well-designed structure, the improvement in the Na^+ storage and cycling stability of SIBs can be achieved [18,25]. For example, Wang et al. prepared multi-layer porous carbon made from activated peanut shells as the anode of SIBs, providing the high capacity up to 154 mAh g^{-1} at 1 A g^{-1} [8,26]. The enhanced electrochemical performance of such anodes for SIBs were attributed to the hierarchically porous morphology of the thrived carbons [26].

In this work, we report the B and P co-doped porous carbon, prepared by a simple one-step pyrolysis method, as the anode materials for SIBs. The sodium-tetraphenylboron ($(\text{C}_6\text{H}_5)_4\text{BNa}$) and chlorodiphenylphosphine ($(\text{C}_6\text{H}_5)_2\text{ClP}$) are applied as B and P sources, respectively. By the advantage of the co-doping of B, P and the porous structure, the obtained material enables high performance SIBs with excellent Na^+ storage, rate and cycling capability. The SIBs can provide reversible discharge/charge specific capacity of 254.90 mAh g^{-1} at 50 mA g^{-1} , and 93.8 mAh g^{-1} after 3000 cycles under the current density of 1 A g^{-1} with the capacity retention up to 90.40%.

2. Experimental

2.1. Preparation of B and P Co-Doped Carbon

Sodium tetraphenylborate ($(\text{C}_6\text{H}_5)_4\text{BNa}$), chlorodiphenylphosphine ($(\text{C}_6\text{H}_5)_2\text{ClP}$) and ethanol ($\text{C}_2\text{H}_6\text{O}$) were purchased from Shanghai Chemical Technology Co., Ltd. The process of B and P co-doped porous carbon started from dissolving 0.02 mol of $(\text{C}_6\text{H}_5)_4\text{BNa}$ in 100 ml of $\text{C}_2\text{H}_6\text{O}$ under vigorous stirring for 5 h, followed by adding 0.02 mol of $(\text{C}_6\text{H}_5)_2\text{ClP}$ dropwise. Thereafter, the mixture was stirred at room temperature for another 6 h till a milky white solution was observed. The resultant solution was centrifuged four times with ethanol to remove excessive Na^+ and Cl^- , and dried in the oven at 70 °C for 12 h. Finally, the obtained white powder was carbonized at 700, 800, 900, and 1000 °C for 3 h at the heating rate of 1 °C min^{-1} and maintained for 3 h under N_2 atmosphere. The obtained materials were ground into fine powder and then stirred with deionized water for 5 h to prepare

homogeneous mixture, which was then filtered with deionized water and dried in the oven at 60 °C for 6 h. The products carbonized at different temperatures were expressed as BPC700, BPC800, BPC900, and BPC1000 for boron, phosphorus co-doped carbon samples synthesized at 700, 800, 900 and 1000 °C, respectively.

2.2. Characterization of materials

The scanning electron microscope (SEM, FEI Quanta 200), transmission electron microscope (TEM, Tecnai F20), and energy-dispersive X-ray spectrometer (EDS) were used to examine the morphology, microstructure, and elemental compositions of the carbon samples. The X-ray diffractometer (XRD, Bede D1, Bede Scientific Ltd., UK) equipped with Cu-K α radiation ($\lambda = 0.15418$ nm) was used to analyze the crystal structure and phase of the material. Raman spectra were obtained by a Jobin Yvon Lab RAM HR800 instrument with an excitation wavelength of 514 nm. X-ray photoelectron spectroscopy (XPS, VG ESCALAB MKII) was used to measure the chemical composition and element bonding of the samples. The Brunauer-Emmett-Teller (BET) specific surface area and the pore size distribution of the samples were examined by Nitrogen adsorption/desorption measurements at 77 K (TriStar II 3020).

2.3. Electrochemical measurements

BPC material, acetylene black and polyvinylidene fluoride (PVDF) were ground (weight ratio, 7: 2: 1) to form homogeneous mixture, which was then added with 1-methyl-2-pyrrolidone (NMP). Next, the slurry was pasted onto the copper foil and dried in a vacuum oven at 60 °C overnight. The dried electrode film was subsequently cut into a disc with a diameter of 13 mm. The mass loading of the active material in working electrodes was between 1.5–2 mg cm^{-2} . The coin cells (CR2032) were assembled in a glove box with Na metal and glass fiber (GF/D, Whatman) as the counter/reference electrode and separator, respectively. The electrolyte was 1.0 M NaClO_4 in ethylene carbonate/diethylene carbonate (1: 1, vol/vol.) with 5 vol% fluoroethylene carbonates added. The electrochemical behaviors of the SIBs were measured with LAND-CT2001A battery test system at different current densities in the voltage range of 0.01 to 3 V.

3. Results and discussion

The schematic diagram of the experimental procedures was exhibited in Fig. 1a and the detail of the synthesis of B and P co-doped porous carbon (BPC) can be found in the experimental section. The precursors mixture of $(\text{C}_6\text{H}_5)_2\text{ClP}$ and $(\text{C}_6\text{H}_5)_2\text{BNa}$ enables the electrostatic attraction between the functional groups of $(\text{C}_6\text{H}_5)_2\text{P}^+$ and $(\text{C}_6\text{H}_5)_4\text{B}^-$, with the formation of NaCl subsequently removed by centrifugation. The phenyl groups were dehydrogenated and polymerized during the carbonization process. The morphology of thus-thrived materials was studied by SEM and TEM. The micro and macro pores of BPC can be found in the SEM images in Fig. 1b-c and Fig.S1a-d, presenting an interesting heterogeneous honeycomb shape with abundant semi-open pores. Accordingly, the TEM and HRTEM images in the Fig. 1d-e and Fig.S1e-f, also confirm the unique porous structure of the as-obtained materials, favorable for Na^+ transport and storage process. The high-resolution transmission electron microscope (HRTEM) image of BPC800 in Fig. 1f shows distinctive turbostratic structure of porous carbon with curved graphite layers in a short-range order. From the corresponding Selected Area Electron Diffraction (SAED) image in Fig. 1g, the scattered diffraction ring is characteristic of amorphous materials with the microstructure of piled layer [27]. The measured interlayer distance (d_{002}) of BPC800 is (0.398 nm), greater than the interlayer distance (0.335 nm) of pure carbon, enabling faster diffusion rate and larger storage capacity of Na^+ . Additionally, EDS mapping images in Fig. 1h-i show the uniform distribution of B, P, and O among

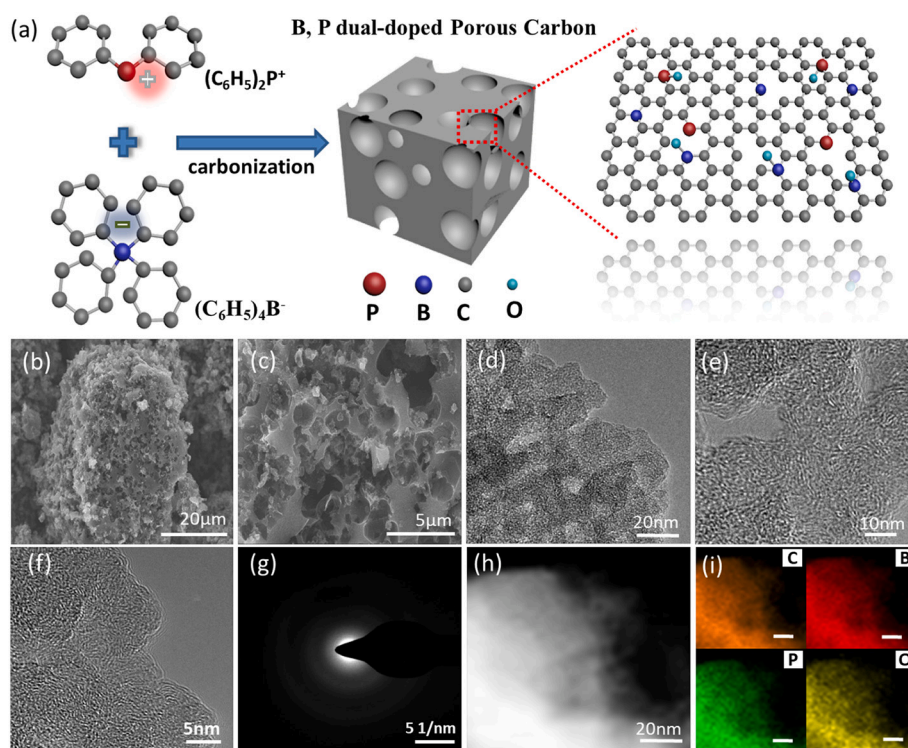


Fig. 1. (a) Schematic illustration of the synthesis of B, P co-doped Porous-Carbon. (b, c) SEM, (d) TEM image, (e, f) HRTEM image, (g) SAED and (h, i) STEM image of BPC800 and the corresponding EDS mapping of C, B, P, and O.

the selected area, indicating the successful doping of B and P in as-prepared samples via the highly feasible procedures.

The XPS analysis was used to study the surface chemistry and element state of BPC samples. As displayed in Fig. S2a, the XPS surveys mainly consist of carbon, phosphorus, boron and oxygen, and the corresponding content of each element is present in Table S1. As shown in Fig. S2b, the C1s deconvolution of BPC samples were divided into five peaks: C–B (284.28 eV), C–C/C=C (284.79 eV), C–P (285.6 eV), C–O (286.6 eV), and O–C=O (289.1 eV) [28,29]. The existence of C–B and C–P bonds directly confirm that the B and P atoms have been successfully merged into the carbon lattice. The high resolution B1s spectra of BPCs at annealing temperatures from 700, 800, 900 and 1000 °C are present in Fig. 2a. Generally, the deconvolution of B1s can be classified into two types: (1) the boron atom replaces the carbon atom inside the carbon skeleton, like BC_3 (189.5 eV) and B_4C (188.0 eV); (2) the boron bonds with oxygen and carbon atoms to form BC_2O (191 eV) and BCO_2 (192.5 eV), located at the edge or defect in the carbon skeleton [28–30]. The B_4C/BC_3 bondings of all samples are less than that of BC_2O and BCO_2 , implying the boron doping tends to exist at the carbon edges or defect sites [30]. Despite the BC_2O and BCO_2 groups are useful to expand the layer space, excessive number of those two groups would cause unavoidable damage to the conductivity of the material [31]. It can be seen from Fig. 2b and Table S2, the percentage of BCO_2 decreases, whilst BC_2O increases as the temperature increases. Contrastingly, the content of BC_3 fluctuates and reaches the maximum at the temperature of 800 °C. Similarly, from the Fig. 2c, P2p spectra are divided into four peaks around 130.50, 134.00, 132.50 and 131.47 eV, corresponding to C_3P , C_2PO_2/CPO_3 , C_3PO and P–O, respectively [28,32,33]. The variation of P-involving groups with the reaction temperature also follows the similar trend as the B-related species (Fig. 2d), where the amount of C–P bonding is the highest at the temperature of 800 °C. It has been reported that the B and P dopant merged in the carbon skeleton can improve the conductivity of the material to enhance the electron/ion diffusion rate.

Furthermore, the heteroatoms located at the edge/defective sites

can expand the layer lattice distance to supply additional reactive sites, as well as present stronger affinity ability of Na^+ to promote storage capacity [20,34].

Additionally, XRD and Raman surveys are utilized to examine the crystalline structure and defect degree of as-obtained BPC materials. The XRD patterns of the BPCs with the carbonization temperatures range of 700, 800, 900 and 1000 °C are shown in Fig. 3a. Two broad peaks near 22° and 43° prove the appearance of (002) and (100) planes, respectively, characteristic of hard carbon materials. Obviously, the (002) peak of BPC800 locates at a relatively smaller angle compared to BPC carbonized at other temperatures. This demonstrates that the BPC800 has the largest interlayer distance (d_{002}) for Na^+ storage compared to the counterparts. In Fig. 3b, all Raman spectra of BPC samples show two independent peaks, sitting at 1335 cm^{-1} (disordered band, D band) and 1584 cm^{-1} (graphite band, G band). The D band normally reveals internally structural defects by detecting the in-plane vibration of sp^2 bonded carbon. The intensity ratio of D-band and G-band (I_D/I_G) is usually applied to evaluate the disorder degree of hard carbon. The I_D/I_G values of BPC carbons decrease along with temperature increasing except BPC at 800 °C, indicating the increment of graphitization degree. The greatest I_D/I_G value of BPC800 indicates abundant defects were generated in the carbon lattices. Generally, it is believed that BPC800 could be the suitable samples as anode for SIB, due to the distinctly greater layer space and defect amount. Fig. 3c shows a N_2 adsorption isotherm and pore size distribution of BPC800, which is characterized by Brunauer Emmett-Teller (BET) gas adsorption measurement. The BET results of BPC700, BPC900 and BPC1000 materials are also analyzed for comparison, as shown in Fig. S2c. The N_2 adsorption isotherm is determined to be type IV isotherm, indicative of considerable mesopores. The surface area of BPC700, 800, 900 and 1000 are 131.4, 212.6, 64.6 and $64.4\text{ m}^2\text{ g}^{-1}$, respectively. The surface area decreases with the pyrolysis temperature except 800 °C, as shown in Table S1. The pore size distribution of BPC800 demonstrates the dominant size range of mesopores is 2–10 nm (inset Fig. 3c), smaller than that of BPC carbonized at other temperature, advantageous for

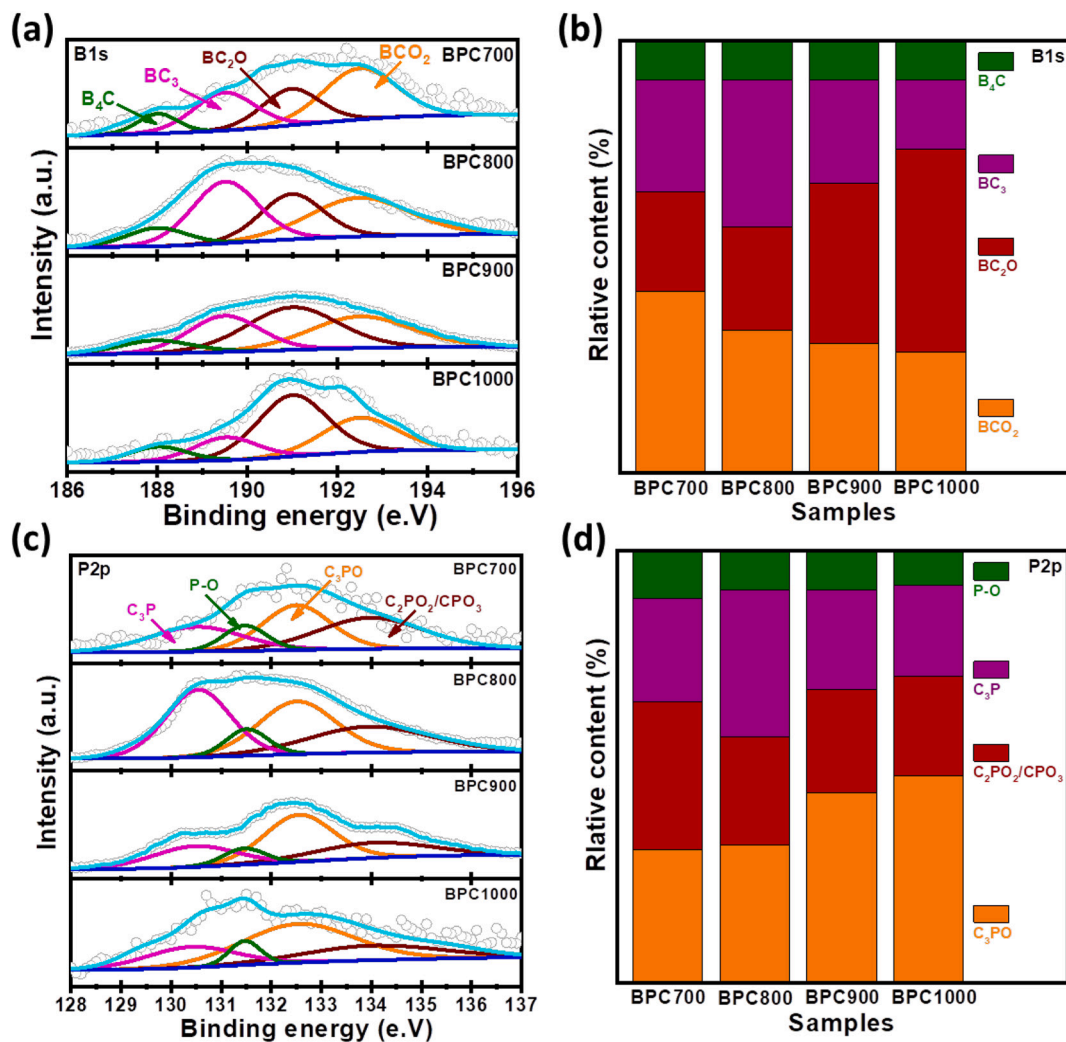


Fig. 2. The deconvolutions of B1s (a) and the content of different B species (b), P2p (c) and the content of P species (d) of BPC700, BPC800, BPC900, and BPC1000 samples from XPS spectra.

Na^+ storage [35–37]. Coupled with SEM, TEM and HRTEM results, it clearly shows the BPC series possess considerable surface area and abundant open pores in the carbon structure, especially BPC800. The larger surface and pore-rich framework enables boosted transport path and broader reaction sites for Na^+ [35,36,38].

The electrochemical analysis of the SIBs assembled with BPC was accomplished via cyclic voltammetry and electrochemical impedance

spectroscopy (EIS). The cyclic voltammograms (CVs) was obtained under the voltage range of 0.01 to 3.0 V (vs Na^+/Na) at the scan rate of 0.1 mV s^{-1} , as shown in Fig. 4a and Fig. S3a–c. In the first negative scan, two irreversible reduction peaks can be observed near 0.0–0.2 and 0.8–1.1 V, ascribed to the formation of the solid electrolyte interface (SEI) film and irreversible side reactions, respectively [8,39,40]. Those two irreversible reduction peaks disappear in the second scan and the

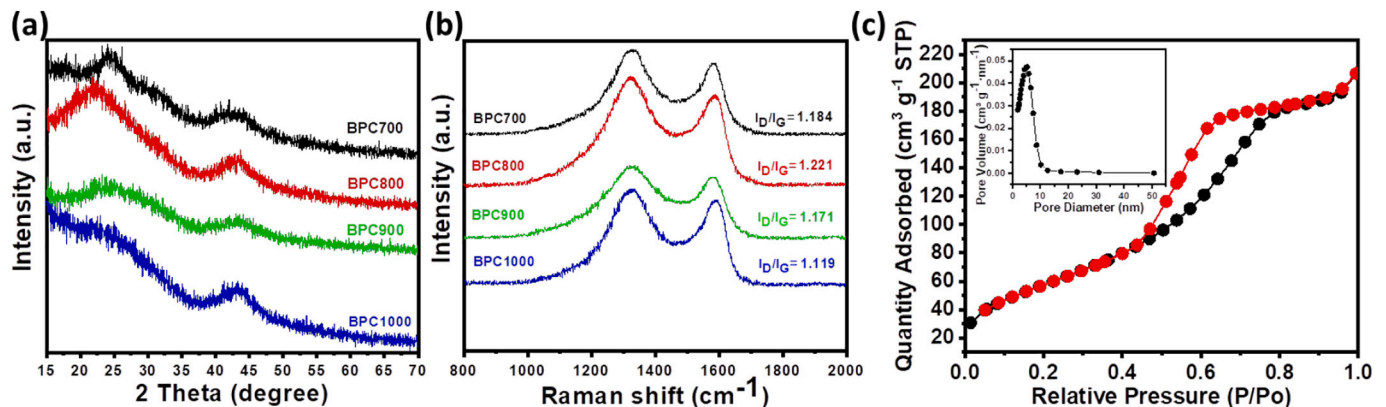


Fig. 3. (a) XRD patterns and (b) Raman spectra of BPC700, BPC800, BPC900 and BPC1000. (c) Nitrogen adsorption/desorption isotherms of BPC800. The inset shows the corresponding pore-size distribution.

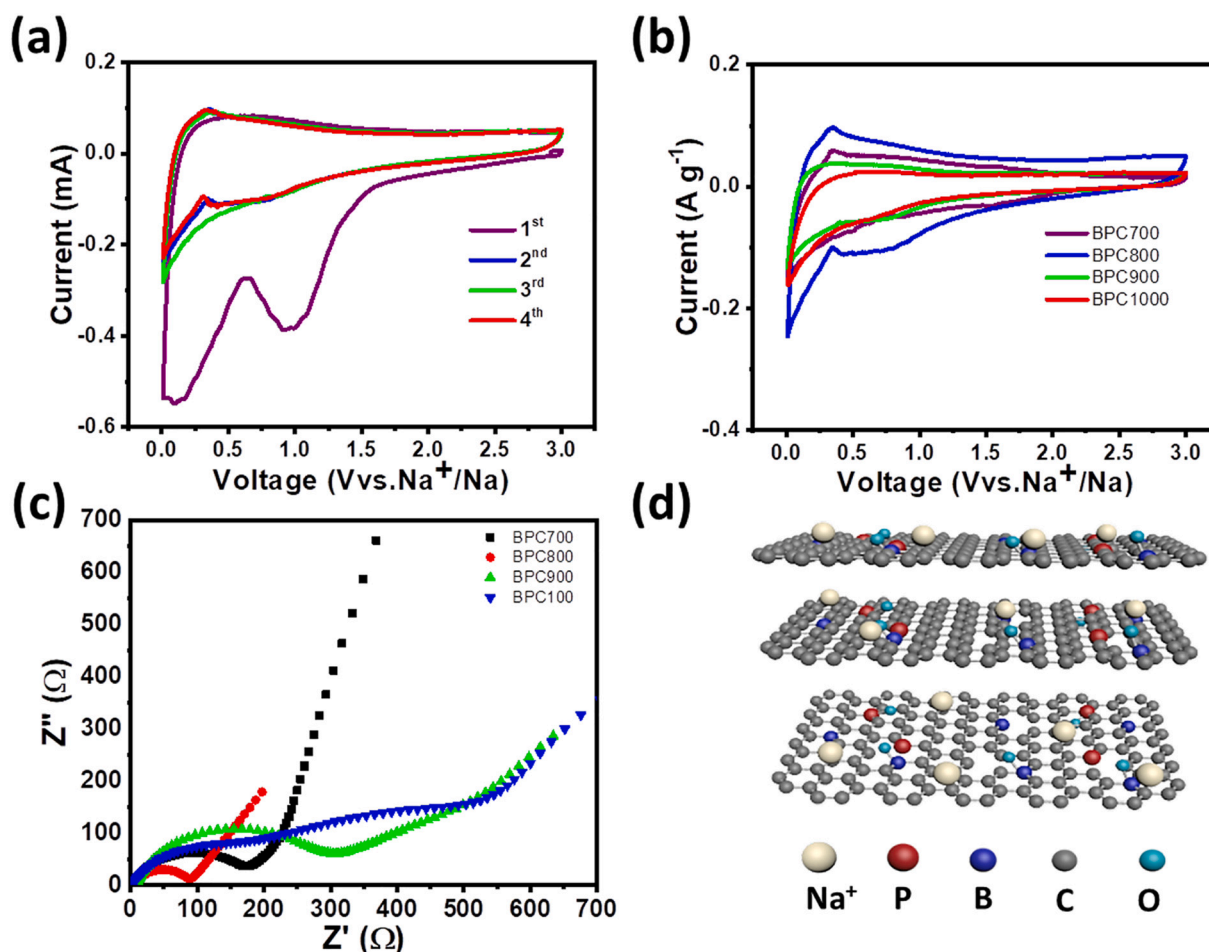


Fig. 4. (a) CVs of BPC800, at a scan rate of 0.1 mV s⁻¹. (b) CVs of BPC700–BPC1000 at scan rate of 0.1 mV s⁻¹. (c) EIS after 1st cycle of BPC700, BPC800, BPC900, and BPC1000. (d) Schematic diagram of Na⁺ diffusion within the porous BPC800 electrode.

subsequent CVs almost overlap, indicating that the stable formation of SEI film after the first cycle [27]. The initial low coulombic efficiency (ICE) occurred due to the formation of SEI by Na⁺ loss and electrolyte decomposition. The SEI film, as cation transfer and electron insulator, can block the path of electrons from electrode to electrolyte and restrain the decomposition of electrolyte, favorable for the lifespan and stability of batteries [41–43]. Due to the Na⁺ insertion between the intermediate layers, a sharp reduction peak is formed in the low potential range (0–0.1 V) of each cycle in all electrodes. There are another pair of important redox peaks around 0.2–0.3 V and 0.8–1.0 V, resulted from the reversible binding of Na⁺ into the defects induced by the heteroatom doping [28,44]. In comparison with BPC700, BPC900 and BPC1000 electrodes (Fig. S3d), the BPC800 electrode has a larger integration area under the peak (Fig. 4b), due to the highly porous structure, larger interlayer space and more defects [8,39]. The results of EIS were quantitatively simulated by the Zview, and the equivalent circuit consisting of R1, R2/CPE1, R3/CPE2 and W (Fig. S4). R1 is the electrode/electrolyte resistance, R2/CPE1 is charge transfer impedance at the interface of electrode/electrolyte in high-frequencies (HF), R3/CPE2 is the charge transfer impedance at medium frequency (MF), and W is the Warburg impedance at the low frequencies (LF) [45–47]. The Nyquist plots BPC at different reaction temperatures are displayed in Fig. 4c. Based on the simulations data, the R1 of BPC700, BPC800, BPC900 and BPC1000 are 4.25, 2.97, 4.18 and 3.70 Ω, respectively (Table S3). BPC800 possesses the lowest electrode resistance, revealing the superior electrical conductivity among the BPCs. This is accordant with previous XPS analysis, as BPC800 owns the largest number of B and P species bonded in the carbon lattice, favorable for conductivity

enhancement. The trend of R2 + R3 among BPC samples follows the similar trends, where the BPC800 presents the minimum value of charge transfer resistance. This phenomenon could probably be resulted from the extended layer space and heterogeneous porous structure of the as-synthesized materials, consequently inducing greater diffusion coefficient of Na⁺ in the BPC800 electrode. The inclined line corresponding to the Warburg diffusion resistance (Z_w) related to the Na⁺ ion diffusion process. Generally, the more vertical line indicates faster ion transport, beneficial for improving electrochemical performance [48]. Clearly, among these electrodes, the slopes of the curve in low-frequency region for BPC700 and BPC800 is relatively more vertical than the counterparts, showing enhanced mass transport capability, guaranteeing the excellent rate and cycle performance. The Nyquist plot of BPCs after the 1st and 10th cycles are shown in Fig. S5 and Table S3. Generally, the R1 and R2 + R3 turn smaller after 10 cycles, possibly because of improved electrolyte penetration in the electrodes during the electrochemical processes. The schematic diagram of mechanism of Na⁺ storage is exhibited in Fig. 4d. On the one hand, the special cross-linked pores and large layer space induced by co-dopants, offer large area and multidirectional routes, guaranteeing the ions diffusion and electrolyte wetting. On the other hand, the dopants in the carbon and adsorption ability of Na⁺. By the advantages of those merits, the porous BPC800 outperformed other BPCs as the anodes for SIBs.

The rate and cycling performance of the BPCs for SIB were examined by Glavano discharge/charge (GCD) techniques. The cycling performance of BPC samples at carbonization temperatures from 700 to 1000 °C were tested at a current density of 0.05 A g⁻¹, as shown in Fig. 5a. The reversible capacities of BPC700, BPC800, BPC900 and

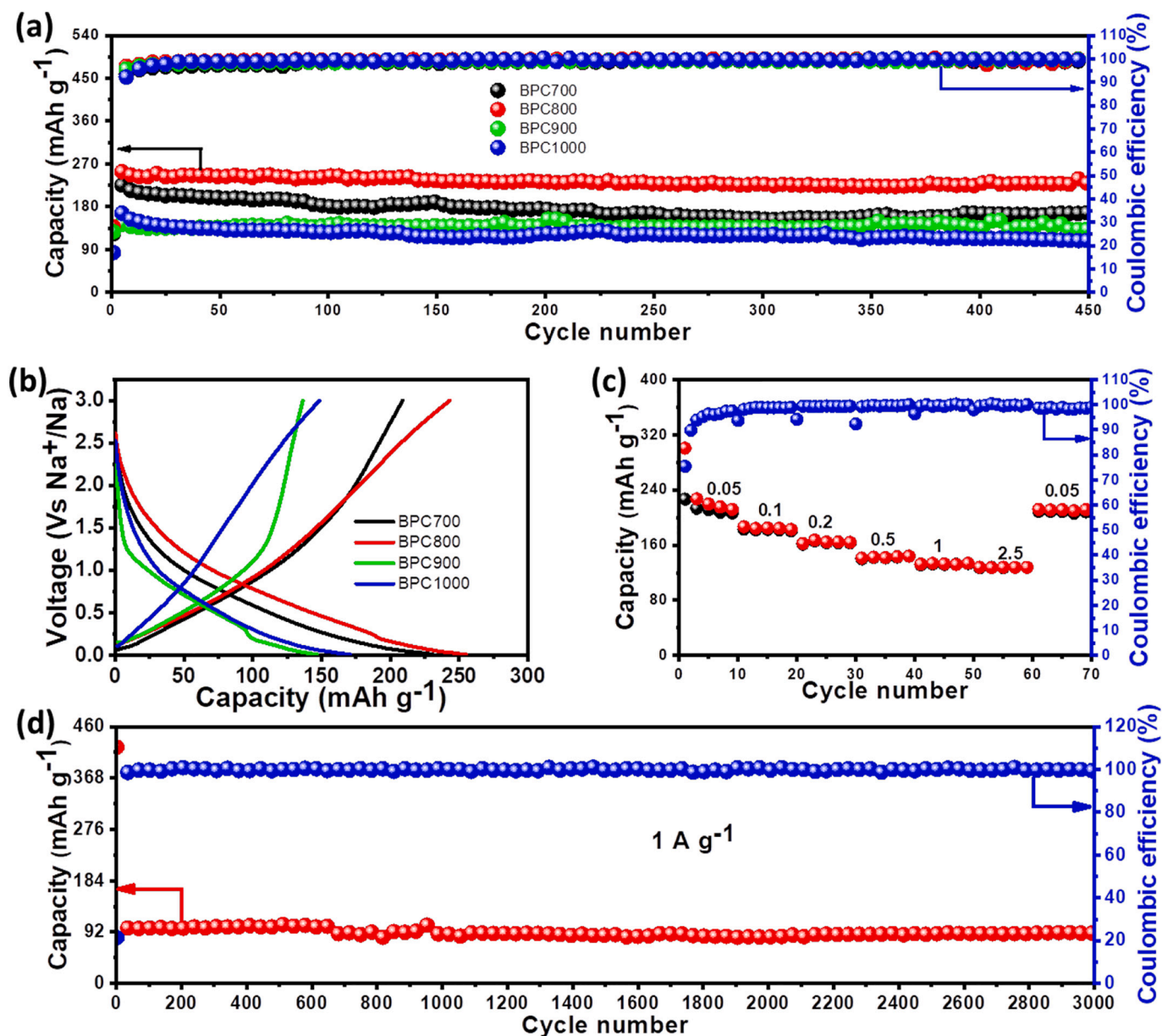


Fig. 5. (a). Cycling performance of BPC700, BPC800, BPC900, and BPC1000 at a current density of 0.05 A g⁻¹. (b) GCD profiles of BPC700, BPC800, BPC900, and BPC1000 at a current density of 0.05 A g⁻¹. (c) Rate performance at different current densities from 0.05 to 1 A g⁻¹ of BPC800. (d) Long-term cycling stability at a current density of 1 A g⁻¹ of BPC800.

BPC1000 after 450 cycles are 173.2, 230.1, 131.9 and 108.2 mA g⁻¹, with the capacity retention of 74.8, 90.3, 89.9 and 63.4%, respectively. It is obvious that BPC800 not only reached the maximum reversible capacity, but also revealed highly stable cycle ability. The superior behavior of BPC800 as the anode for SIB could be attributed to the expanded interlayer distance and abundant defects, providing sufficient space for faster Na⁺ transport and interactions, and consequently leading to greater Na⁺ storage capability. Fig. S6a, displays the cycling behavior of BPC800, harvesting 187.2 mAh g⁻¹ at 0.05 A g⁻¹ after 1050 cycles with the capacity retention of 73.77%. The scale of discharge capacity lies in the insertion and redox process of Na⁺ in carbon structures. The GCD profiles of all samples can be divided into several plateaus: (1) Na⁺ intercalation between the carbon layer (< 0.1 V); (2) Na⁺ binding with defects and pores (> 0.1 V) [28,40,44,49]. The distributions of charge storage in GCD curves are in accordance with CVs (Fig. S6b). The discharge capacity achieved 245.3 mAh g⁻¹ after 80 cycles with the columbic efficiency (CE) of 99.6%. The discharging

curves of BPC samples for SIBs under the current density of 0.05 A g⁻¹ are shown in the Fig. 5b. The discharge capacities of BPC700-BPC1000 are 231.4, 259.8, 146.5 and 170.6 mAh g⁻¹, respectively. Compared to BPC700, BPC900 and BPC1000, BPC800 has a higher reversible capacity. It has been reported that the sodium metal plating and polarization occur at low voltage or high current densities [50]. The CV curves of BPC800 has impressive slope-leading and reversible features, thus greatly mitigating the risk of Na plating and over-polarization during the electrochemical process. The rate performance of the BPC800 is present in Fig. 5c, where the reversible capacity is 247.5 mAh g⁻¹ at the current density of 0.05 A g⁻¹. When the current density increases from 0.1, 0.2, 0.5 and 1 A g⁻¹, the specific capacity gradually decreases as 186.7, 167.4, 142.6, and 127.2, mAh g⁻¹, respectively. The SIB enables the capacity of 222 mAh g⁻¹ as the current density returned to 0.05 A g⁻¹, close to the initial reversible capacity. The cycling performance at high current density of 1 A g⁻¹ is shown in Fig. 5d, where the BPC800 electrode enabled the reversible capacity of 93.8 mAh g⁻¹

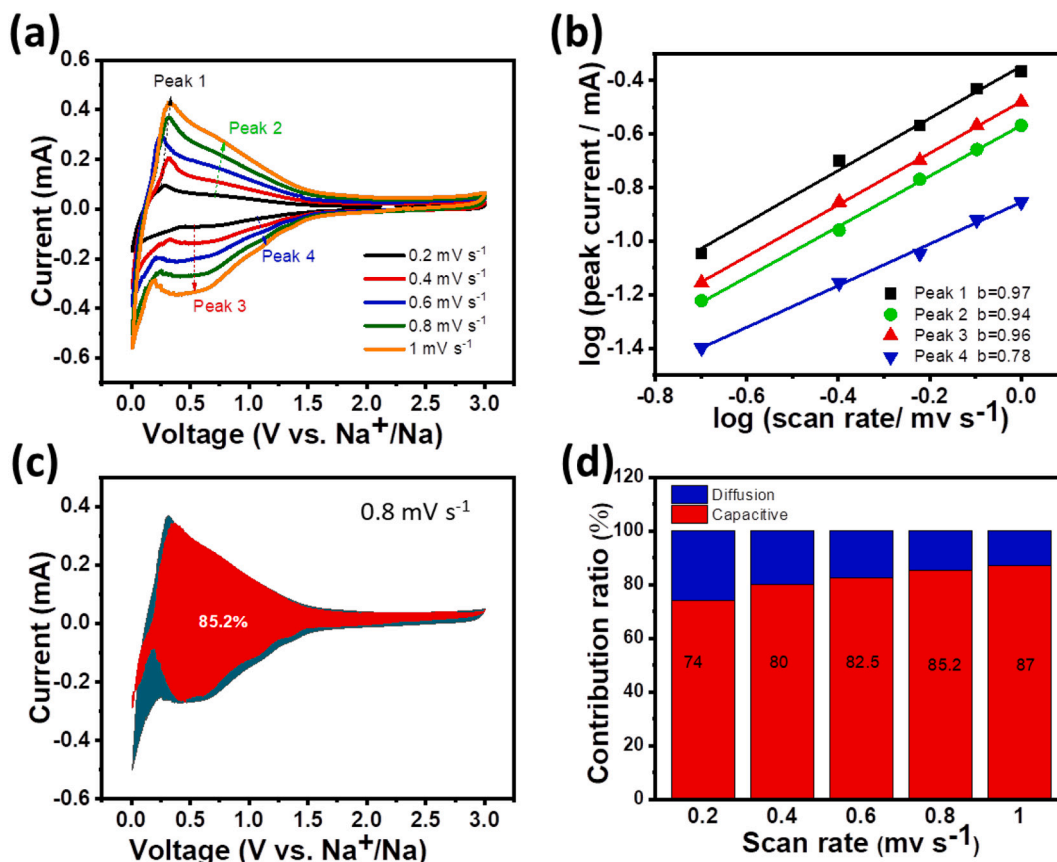


Fig. 6. (a) CV curves of BPC800 at different scan rates after cycling. (b) Relationship between $\log(i)$ and $\log(v)$ for BPC800. (c) CV curve with capacitive- and diffusion-controlled contributions at 0.8 mV s^{-1} and, (d) Bar chart showing the contribution ratio of capacitive- and diffusion-controlled capacities at different scan rate.

after 3000 cycles with superior capacity retention of 90.4%. The excellent rate performance and cyclability can be mainly credited to the heterosporous morphology, effectively buffering the volume change during the repeated insertion/extraction processes [39,51]. Overall, rich surface group and larger inter layer induced co-dopants of B and P, coupled with the 3D porous structure, are beneficial to Na⁺ interaction and storage in SIBs [20,52].

The CV curves at various scan rates are performed to study the storage mechanism and electrochemical reaction process of sodium (Fig. 6a). Generally, the storage mechanism of Na⁺ involves two types: (1) capacitive process, including the adsorption and redox reaction of Na⁺ on/with surface functional groups, edges and defects; (2) diffusion process, involving Na⁺ insertion and extraction. The correlation between peak current (i) and scan rate (v) can be summarized as shown in Eq1 [53].

$$i = av^b \quad (1)$$

$$\log i = b \log v + \log a \quad (2)$$

where i and v represent current density and scan rate, and a and b are constants respectively. Under the condition of b close to 0.5, it means the reaction is controlled by diffusion, whilst if $b = 1$, it implies capacitive process is dominant. The b value obtained for graph of the relationship between $\log I$ and $\log v$ using Eq.2. As shown in Fig. 6b, the b values of peaks 1, 2, 3 and 4 are 0.97, 0.94, 0.96 and 0.78, respectively. These high b value indicate that the storage capacity of the BPC800 electrode mainly depends on the capacitive contribution. Moreover, the total contribution of BPC800 electrodes, including the capacitor-like (k_1v) and diffusion control ($k_2v^{1/2}$) storage, is distinguished under a fixed potential (V) according to Eq.3 [53,54].

$$i(V) = k_1v + k_2v^{1/2} \quad (3)$$

where k_1 and k_2 are constants. Eq. 3 can be rewritten as Eq. 4:

$$i(V)/v^{1/2} = k_1v^{1/2} + k_2 \quad (4)$$

The values of k_1 and k_2 can be calculated by drawing the fitted lines of $i(V)/v^{1/2}$ and $v^{1/2}$, and then the capacitance current $i_c(V) = k_1v$ can be distinguished from the total measured current. As shown in Fig. 6c, the red area shows the proportion of capacitive-process to the current, which is 85.2% of the ultimate capacity under the scan rate of 0.8 mV s^{-1} . Additionally, the capacitance contribution of BPC800 at scan rates of 0.2, 0.4, 0.6 and 1.0 mV s^{-1} , and the corresponding capacitance contribution rates are 74%, 80%, 82.5% and 87%, respectively (Fig. 6d). The contribution of capacitive capacity is on the rise as the scan rate increases probably due to the fast electron transfer, favorable for the high rate capability. The dominant capacitive behavior enables large amount of Na⁺ easily stored on the surface of the material. Hence, the superior adsorption capability of B, P co-doped carbons consequently leads to the excellent rate performance and cycle stability [53,55].

4. Conclusion

To conclude, one-step synthesis of B and P co-doped porous carbon (BPC) was achieved through the pyrolysis of inexpensive precursors of sodium-tetraphenylboron and chlorodiphenylphosphine. Thus-obtained BPC800 harvests the high capacity of 245.0 mAh g^{-1} after 100 cycles and maintains 187.2 mAh g^{-1} after 1050 cycles at 0.05 A g^{-1} . When the applied current was increase to 1 A g^{-1} , the SIB with BPC800 achieves 93.8 mAh g^{-1} after 3000 cycles, with remarkable capacity retention of 90.4%. These results indicate that the porous morphology

of BPC800 can promotes charge transfer, ion diffusion and therefore the SIB with BPC800 anode has a better rate capability. The outstanding electrochemical performance of BPC carbons are attributed to the significant increase in conductivity, defects, and interlayer distance, induced by successful co-doping of B and P in the carbon lattice, as well as the favorable porous structure. Besides, the SIBs with BPC carbons enable to alleviate the volume variation along with the repetitive insertion and extraction. This work offers a simple and facile approach to prepared carbons with dual dopants (B and P) for promising anodes for SIBs.

Credit author statement

Nazir Ahmad: Data curation, Writing-Original draft preparation. Majid Khan: Investigation. Xiangjun Zheng: Visualization. Zhihui Sun: Investigation. Jin Yan: Visualization. Chaohui Wei: Reviewing and Editing. Liwei Shen: Validation. Nadia Batool: Validation. Ruizhi Yang: Conceptualization, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ssi.2020.115455>.

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